LAB 4: ACTIVITY DIAGRAM - CHECKOUT & ORDER FULFILMENT

Submitted by: Sparsh Tyagi, Vaishnavi Rajawat, Shubham Patel, Varnit Tomar **Faculty Mentor:** Mr. Rishabh Rathore Sir

AIM

To construct the UML Activity Diagram with Swimlanes modeling the control flow of the multi-vendor checkout, inventory verification, payment authorization, and vendor order splitting workflow.

THEORY & SWIMLANE ALLOCATION

An Activity Diagram depicts the sequence of execution steps and decision logic. Swimlanes partition responsibilities across three system actors:

- **Customer Swimlane:** Initiates cart checkout, enters delivery address, applies coupons, and authorizes payment.
- **System / API Gateway Swimlane:** Validates stock availability, computes totals, interacts with the Payment Gateway, and splits parent orders into vendor sub-orders.
- **Vendor Swimlane:** Receives assigned sub-orders, packs inventory, updates shipment status, and completes fulfilment.

ACTIVITY WORKFLOW DIAGRAM

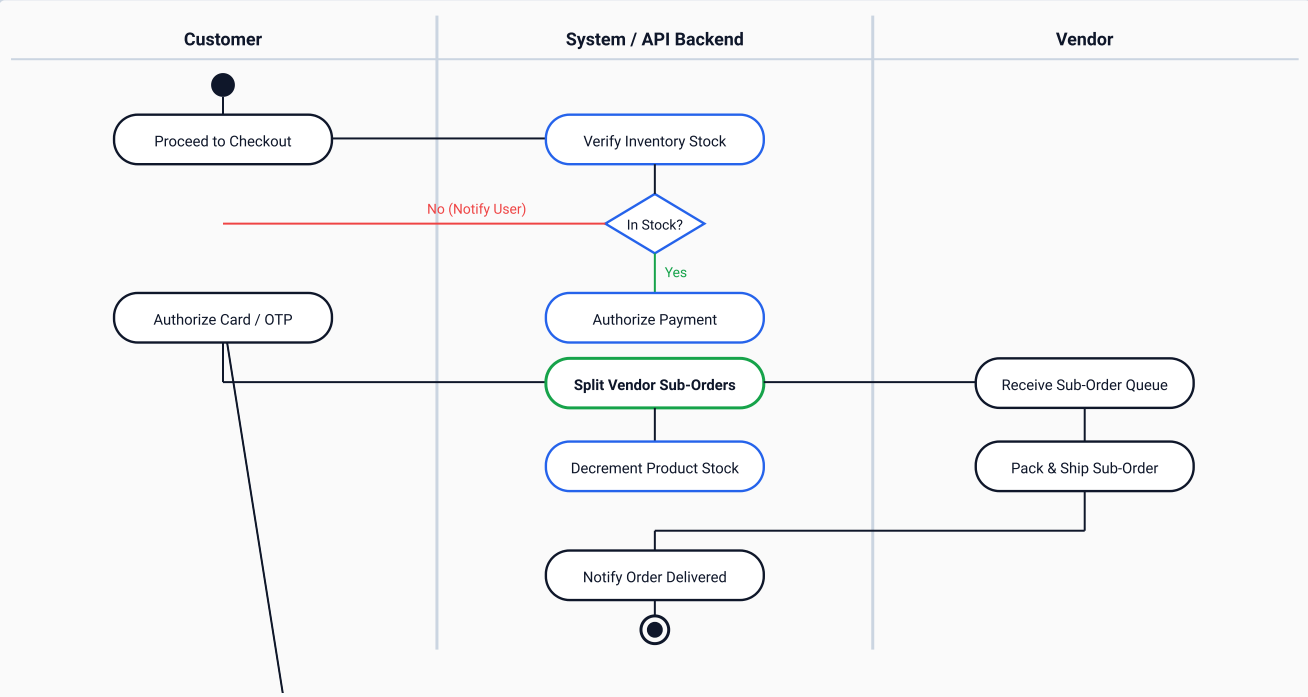


FIGURE 4.1: ACTIVITY DIAGRAM WITH SWIMLANES FOR CHECKOUT & MULTI-VENDOR ORDER PROCESSING

CONCLUSION

The Activity Diagram clearly delineates the control flow across Customer, System API, and Vendor swimlanes, illustrating inventory locking, payment validation, and multi-vendor sub-order distribution.